

Chap 2

Derivatives

* The Chain Rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

- For a given function $y = f(g(x))$, let $u = g(x)$.
Then $y = f(u)$ and $u = g(x)$.
By the chain rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= f'(u) \cdot g'(x) \\ &= f'(g(x)) \cdot g'(x)\end{aligned}$$

* Trigonometric Function's Differentiation Rules

- (1) $y = \sin x \Rightarrow y' = \cos x$
- (2) $y = \cos x \Rightarrow y' = -\sin x$
- (3) $y = \tan x \Rightarrow y' = \sec^2 x$
- (4) $y = \cot x \Rightarrow y' = -\csc^2 x$
- (5) $y = \sec x \Rightarrow y' = \sec x \tan x$
- (6) $y = \csc x \Rightarrow y' = -\csc x \cot x$

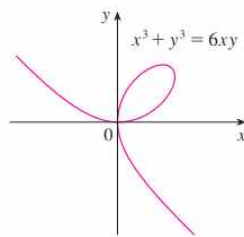
§6 Implicit Differentiation

* Representation for Function

$$(1) \ y = f(x): \ y = \sqrt{x^3 + 1}, \quad y = x \sin x$$

$$(2) \ F(x, y) = 0: \ x^2 + y^2 = 25, \quad x^3 + y^3 = 6xy$$

* The equation $x^3 + y^3 = 3axy$ (a : constant) is called the **folium of Descartes**. (데카르트의 정엽선)



• The parametric equation of the folium of Descartes $x^3 + y^3 = 3axy$

$$\text{are } x = \frac{3am}{1+m^3}, \quad y = \frac{3am^2}{1+m^3}.$$

(pf) Let $y = mx$

$$\Rightarrow x^3 + m^3 x^3 = 3amx^2$$

$$\Rightarrow x^2 \{ (1+m^3)x - 3am \} = 0$$

$$\Rightarrow x = \frac{3am}{1+m^3}, \quad y = \frac{3am^2}{1+m^3}$$

Exam 1 (a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

(sol) (a) $x^2 + y^2 = 25$

$$\Rightarrow \frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\Rightarrow \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

$$\Rightarrow 2x + \frac{d}{dy}(y^2) \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

(b) The slope of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ is

$$\left. \frac{dy}{dx} \right|_{(3, 4)} = -\left. \frac{x}{y} \right|_{(3, 4)} = -\frac{3}{4}.$$

$\therefore$ The equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ is

$$y - 4 = -\frac{3}{4}(x - 3) \text{ or } 3x + 4y = 25.$$

Exam 2 (a) Find y' if $x^3 + y^3 = 6xy$.

(b) Find the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

(c) At what point in the first quadrant is the tangent line horizontal?

(sol) (a) $x^3 + y^3 = 6xy$

$$\Rightarrow 3x^2 + 3y^2 y' = 6y + 6x y'$$

$$\Rightarrow 3(y^2 - 2x) y' = 3(2y - x^2)$$

$$\Rightarrow y' = \frac{2y - x^2}{y^2 - 2x}$$

(b) The slope of the tangent is $y'|_{(3, 3)} = \frac{6-9}{9-6} = -1$.

$\therefore$ The equation of the tangent to the folium at $(3, 3)$ is

$$y - 3 = -(x - 3) \text{ or } y = -x + 6$$

(c) The tangent line is horizontal if $y' = 0$.

$$\Rightarrow y' = \frac{2y - x^2}{y^2 - 2x} = 0$$

$$\Rightarrow y = \frac{x^2}{2}, \quad x^3 + y^3 = 6xy$$

$$\Rightarrow x^3 + \frac{x^6}{8} = 6x \frac{x^2}{2}$$

$$\Rightarrow x^3(x^3 - 2^4) = 0$$

$$\Rightarrow (x, y) = (0, 0), \quad (2^{4/3}, 2^{5/3})$$

$\therefore$ The tangent line is horizontal at $(2^{4/3}, 2^{5/3})$ in the first quadrant.

Exam 3 Find y' if $\sin(x+y) = y^2 \cos x$.

(sol) $\sin(x+y) = y^2 \cos x$

$$\Rightarrow \cos(x+y)(1+y') = 2y(\cos x) y' - y^2 \sin x$$

$$\Rightarrow y' \{2y \cos x - \cos(x+y)\} = \cos(x+y) + y^2 \sin x$$

$$\Rightarrow y' = \frac{\cos(x+y) + y^2 \sin x}{2y \cos x - \cos(x+y)}$$

Exam 4 Find y'' if $x^4 + y^4 = 16$.

(sol) $x^4 + y^4 = 16$

$$\Rightarrow 4x^3 + 4y^3 y' = 0$$

$$\Rightarrow y' = -\frac{x^3}{y^3}$$

$$\begin{aligned} \text{I. } y'' &= -\frac{3x^2 y^3 - x^3 3y^2 y'}{y^6} \\ &= -\frac{3x^2 y^3 - 3x^3 y^2 \left(-\frac{x^3}{y^3}\right)}{y^6} \\ &= -\frac{3x^2 y^4 + 3x^6}{y^7} \\ &= -\frac{3x^2 (y^4 + x^4)}{y^7} \\ &= -\frac{48x^2}{y^7} \end{aligned}$$

II. $4x^3 + 4y^3 y' = 0$

$$\Rightarrow x^3 + y^3 y' = 0$$

$$\Rightarrow 3x^2 + 3y^2 (y')^2 + y^3 y'' = 0$$

$$\Rightarrow 3x^2 + 3y^2 \left(-\frac{x^3}{y^3}\right)^2 + y^3 y'' = 0$$

$$\Rightarrow 3x^2 y^4 + 3x^6 + y^7 y'' = 0$$

$$\begin{aligned} \Rightarrow y'' &= -\frac{3x^2 (y^4 + x^4)}{y^7} \\ &= -\frac{48x^2}{y^7} \end{aligned}$$

Exercise 22 Find an equation of the tangent line to the curve $x^{2/3} + y^{2/3} = 4$ at the given point $(-3\sqrt{3}, 1)$.

(sol) $x^{2/3} + y^{2/3} = 4$

$$\Rightarrow \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0$$

$$\Rightarrow y' = -\frac{y^{1/3}}{x^{1/3}}$$

$\therefore$ The slope of the tangent to the curve at the point $(-3\sqrt{3}, 1)$ is

$$y'|_{(-3\sqrt{3}, 1)} = -\frac{1}{\sqrt[3]{-3\sqrt{3}}} = \frac{1}{\sqrt{3}}.$$

$\Rightarrow$ The equation of the tangent line is

$$y - 1 = \frac{1}{\sqrt{3}}(x + 3\sqrt{3}) \quad \text{or} \quad x - \sqrt{3}y + 4\sqrt{3} = 0.$$